

Practice Exam 1

Score

MTH 207 (Fall 2026) — J. Baggett

/ 132

First Name	Last Name

Before exam begins:

- Write your first name and last name above.
- Turn off all electronics and keep them out of sight: no cellular phones, not even to tell time (and not even if it's just in airplane mode). No iPods, wearing of headphones, etc.

As soon as exam starts:

- Take a quick breath to relax!!
- Check that you have all 9 pages of the exam. (The number of pages includes this cover page.)

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
 - Explain your reasons and justify your steps.
 - Demonstrate that you understand the *process*.
 - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.

The real exam is 55 minutes and 100 points. This practice exam is deliberately longer (120 points) so it covers more of the material — do not use it to judge timing.

- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

(3^{pts} ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) If the average rate of change of f on $[1, 3]$ equals the average rate of change on $[3, 5]$, then f is linear on $[1, 5]$.

(b) If $\lim_{x \rightarrow 5} f(x) = 2$ and $\lim_{x \rightarrow 5} g(x) = 0$, then $\lim_{x \rightarrow 5} \frac{f(x)}{g(x)}$ does not exist.

(c) The graph of $y = \frac{3x}{\sqrt{x^2+1}}$ approaches the same horizontal line as $x \rightarrow \infty$ and as $x \rightarrow -\infty$.

(d) If $s(t)$ is position, then $s'(a)$ is the instantaneous velocity at time a , and it can be negative.

- (8^{pts}) **2.** Let $f(x) = x^2 - 3x$. Find the slope of the secant line through the points $(1, f(1))$ and $(1+h, f(1+h))$, and simplify it. What value does that slope approach as $h \rightarrow 0$, and what is that value called?

(18^{pts}) **3.** Evaluate the following limits. As a reminder, **justify your steps!** Simplify your answer as much as possible. (If the limit does not exist, state why.)

- $\lim_{x \rightarrow -1} (2x^2 - 5x + 4)$

- $\lim_{t \rightarrow 3} \frac{t^2 + t - 12}{t - 3}$

- $\lim_{x \rightarrow 0} x^4 \cos\left(\frac{1}{x}\right)$

(18^{pts}) **4.** Evaluate the following limits. As a reminder, **justify your steps!** (If the limit does not exist, state why.)

- $\lim_{x \rightarrow 1} \frac{x - 1}{|x - 1|}$

- $\lim_{x \rightarrow 0} \frac{\cos x}{2x + \sin x}$

- $\lim_{x \rightarrow 2} e^{x^2 - 3x}$

(8^{pts}) **5.** Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$. (Hint: rationalize the numerator.)

(7^{pts}) **6.** State the Squeeze Theorem.

(10^{pts}) **7.** Let $f(x) = \frac{x+1}{x-3}$. Find $\lim_{x \rightarrow 3^+} f(x)$ and $\lim_{x \rightarrow 3^-} f(x)$, and say what this tells you about the graph of f .

(11^{pts}) **8.** Find the horizontal asymptotes of $f(x) = \frac{3x^2 + 8}{7x^2 + 5}$.

(8^{pts}) **9.** Evaluate the following limit or state that it does not exist. Show the steps!

$$\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + 2x} - x \right)$$

(15^{pts}) **10.** Let $f(x) = 3 + \frac{4}{x}$. Use the **definition of derivative** to compute $f'(5)$. Note that using the Power Rule $\frac{d}{dx}(x^n) = nx^{n-1}$ will not earn points.

(3^{pts}) **11.** Show by means of an example that $\lim_{x \rightarrow a} [f(x) + g(x)]$ may exist even though neither $\lim_{x \rightarrow a} f(x)$ nor $\lim_{x \rightarrow a} g(x)$ exists.

(4^{pts}) **12.** Give a formula for a function that is continuous at $x = 2$ but **not** differentiable at $x = 2$. Briefly say why your example works.

(10^{pts}) **13.** Find the x -coordinate(s) of any point(s) where the tangent line to the curve $y = x^4 + 1$ is parallel to the line $32x - y = 15$.

THIS IS THE LAST PAGE OF THE PRACTICE EXAM.